

Deterministic Optimization

Geometric Aspects of Linear Optimization

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Extreme Points and Convex Hull

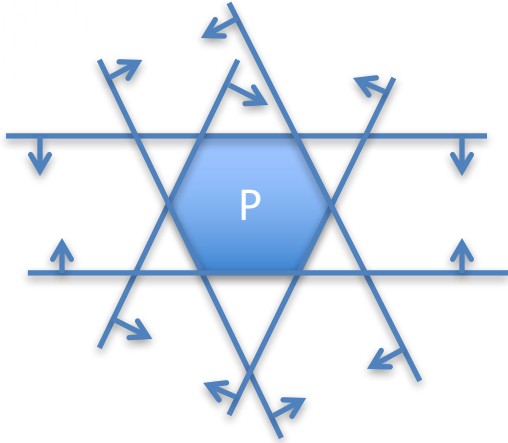
Geometric Aspects of Linear Optimization

Learning Objectives for this lesson

- Discover an alternative way to represent a polyhedron
- Study key geometric objects and operations:
 - Extreme points
 - Convex combination
 - Convex hull

Halfspace Representation of Polyhedron

- Definition: A **polyhedron** is the **intersection of a finite number of halfspaces**

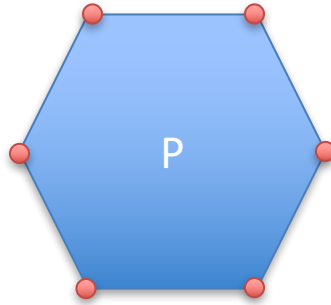
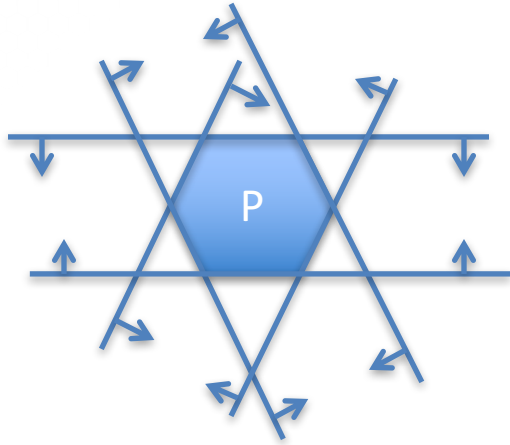


$$P = \{x \in R^n \mid a_1^\top x \leq b_1, \\ \dots \\ a_m^\top x \leq b_m\}$$



Another Representation of Polyhedron

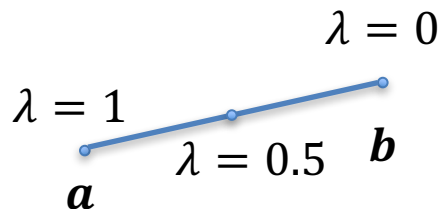
- Instead of edges, look at “Corner Points”



- “Corner points” are responsible for “generating” the set

Convex Combination of Two Points

- Given two points $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, a convex combination of $\mathbf{a}, \mathbf{b}$ is given by
 - $\mathbf{x} = \lambda \mathbf{a} + (1 - \lambda) \mathbf{b}$ for some $\lambda \in [0, 1]$.
 - Geometrically, $\mathbf{x}$ is on the *line segment* connecting $\mathbf{a}$ and $\mathbf{b}$

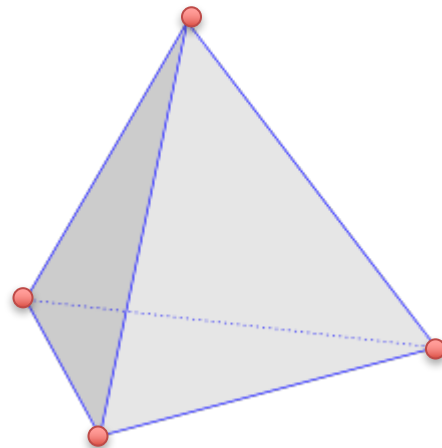
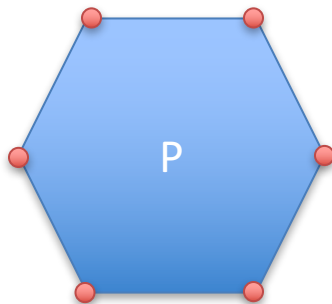
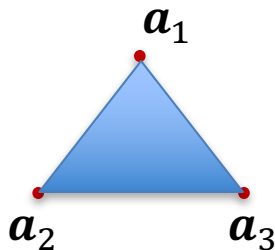


Convex Combination of m points

A point x is a convex combination of $a_1, \dots, a_m$ if x can be written as

- $x = \sum_{i=1}^m \lambda_i a_i$
- And $\sum_{i=1}^m \lambda_i = 1, \lambda_i \geq 0$ for $i = 1, \dots, m$.

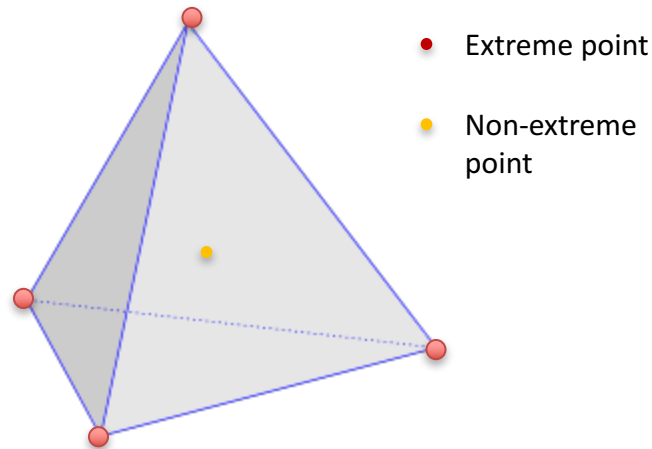
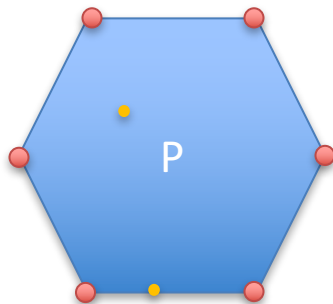
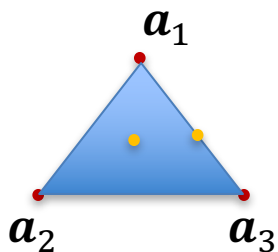
- Examples:



- “Corner points” are special points. We give them a special name.

Extreme Point

- A point x in a polyhedron P is an **extreme point** if and only if x is not a convex combination of other two different points in P .
 - There do not exist $y, z \in P$, such that $y \neq x, z \neq x$ and $x = \lambda y + (1 - \lambda)z$ for some $\lambda \in [0, 1]$.
- Extreme points are “corner points”.



Convex Hull and Bounded Polyhedron

- A **convex hull** of m points $a_1, \dots, a_m$ is the set of all convex combinations of $a_1, \dots, a_m$, denoted as $\text{conv}\{a_1, \dots, a_n\}$.
 - $\text{conv}\{a_1, \dots, a_n\} = \{x \mid x = \lambda_1 a_1 + \dots + \lambda_n a_n, \lambda_1 + \dots + \lambda_n = 1, \lambda_i \geq 0 \forall i = 1, \dots, n\}$
- **Theorem:** A **nonempty** and **bounded** polyhedron is the **convex hull** of its **extreme points**.
- A bounded polyhedron is a polyhedron that does not extend to infinity in any direction.
- A bounded polyhedron is also called a **polytope**.

Summary

- We learned:
 - Convex combination
 - Convex hull
 - Extreme points
 - Polytope as convex hull of extreme points